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Physiological screening for drought-tolerance traits among hemp (*Cannabis sativa* L.) cultivars in controlled environments and in field

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ABSTRACT

The identification of genetic traits that enable water conservation could improve water availability for plant use when drought may develop. Hemp (Cannabis sativa L.) production in the USA has continued to increase dramatically after the federal government legalized its cultivation in 2014, but because of strict regulations in the past, necessary research for identifying adapted cultivars with desirable traits is lacking. The objectives of this study were to identify water-saving traits among hemp cultivars and determine if they influence cannabidiol (CBD) potency. Three approaches were taken: (i) hemp cultivars transpiration rate (TR) responses to soil drying were evaluated in a greenhouse, (ii) Sensitivity of cultivars TR to high vapor pressure deficit (VPD) was tested in a walk-in chamber, (iii) Field trial data were examined to determine if expression of drought-tolerance traits was associated with changes in CBD potency. Five out of thirteen cultivars had the earliest stomatal closure at rates higher than fraction of transpirable soil water (FTSW) threshold of 0.55. Four out of twelve expressed the limited transpiration rate (TRlim) trait, with their VPD thresholds ranging from 1.2 to 2.6 kPa. Cultivar Ha3ze showed a desired early decrease in TR during soil drying, delayed wilting in the field when soil-water deficit developed, and had greater CBD percentage than other cultivars. This study provides useful information on variability among hemp cultivars in TR response to soil drying and high evaporative demand that can be applied in developing cultivars that are better suited to a range of water-limited conditions.

Abbreviations: CBD, Cannabidiol; Cherry NT, Cherry Not Topped; Cherry T, Cherry Topped; DAT, Days after transplant; FTSW, Fraction of transpirable soil water; RWC, Relative water content; SLA, Specific leaf area; T1 NT, T1 Not Topped; TRlim, Limited transpiration rate; TR, Transpiration Rate; T1 T, T1 Topped; VPD, Vapor pressure deficit; WS, Wilting score

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Hemp; drought; fraction of transpirable soil water; CBD; limited transpiration; vapor pressure deficit

1. Introduction

Hemp (*Cannabis sativa* L.) production in the USA has increased tremendously because of the passing of the law regarding its cultivation in 2014, and

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the 2018 Farm Bill, which identified hemp as an agricultural commodity. In 2019, 206,965 hectares were licensed for growing hemp in the USA, which was a 554% increase from 31,637 hectares in 2018 (Angell 2019).

As a multipurpose crop with three harvestable parts: grain, fiber, and flower, the versatility of this plant boosts its popularity, and the potential of new markets is appealing to farmers. While traditionally a fiber and grain crop, hemp has emerged as a source of non-hallucinogenic medicinal phytocannabinoids (e.g., CBD) (Schluttenhofer and Yuan 2017). Because oilseed cultivars produce more flowers than fiber cultivars, and the inflorescences are where most cannabinoids are produced, these cultivars are a logical source for CBD. Moreover, the use of industrial hemp for the production of CBD has not received much consideration, although there is commercial interest in using field-grown plants for producing CBD (Cherney and Small 2016).

The most important physiological process determining crop resource-use efficiency is photosynthesis (Tang et al. 2017). However, transpiration is a natural consequence of stomatal opening to allow CO_2 to diffuse into the interior of the leaf to be assimilated in photosynthesis. Hence, transpirational water loss is necessary and quantitatively linked to plant growth and production (Tanner and Sinclair 1983).

To overcome the negative impacts of drought stress in crops, many strategies have been investigated and adopted, mainly including agronomic management and genetic improvement (Kebede et al. 2014). The physiological traits that appear to be particularly desirable to improve performance of crops under water-deficit conditions are generally associated with limited water use early in the growing season so that water is conserved for use later in season to sustain plant productivity during seed fill (Manandhar et al. 2017; Shekoofa and Sinclair 2018).

Studies have shown that partial stomatal closure in response to both progressive soil drying and high atmospheric vapor pressure deficit (VPD) can confer water-saving properties on plants because of reduced plant transpiration rate (TR) (Shekoofa et al. 2013; Devi et al. 2014; Sinclair et al. 2018). Therefore, plants expressing one or both stomatal closure traits can conserve soil water and prolong crop physiological activity under dry conditions.

The first response to a leaf water deficit is stomatal closure (Hura et al. 2007), which occurs before any change in relative water content (RWC) occurs. According to Wanjiku et al. (2019), RWC can be used to monitor water status of the plant during growth and under drought stress. RWC is percent of water at a given time as related to the water content at full turgor (Gonzalez and Gonzalez-Vilar 2001).

While substantial progress has been made in identifying varieties of peanut (*Arachis hypogaea*), maize (*Zea mays*), sorghum (*Sorghum bicolor*), cowpea (*Vigna unguiculata*), and soybean (*Glycine max*) that possess drought-tolerance traits, little or no attention has been given to hemp cultivars. Since

the production of hemp in the USA is rapidly increasing, transpiration rate (i.e., water loss) under varying environments (i.e., water deficit conditions and high evaporative demand), as a consequence of photosynthesis, should be assessed among hemp cultivars.

The first objective of this study was to identify the water-saving traits, including fraction of transpirable soil water (FTSW) threshold and limited transpiration (TR_{lim}) among hemp cultivars in controlled environments. The second objective was to confirm if the water-saving potential of those cultivars that expressed one or both drought-tolerance traits can improve CBD concentration compared to cultivars that did not express the drought-tolerance traits in the field, under rainfed conditions.

2. Materials and methods

In two controlled-environment studies [transpiration rate (TR) response to progressive soil-drying and TR response to varying evaporative demand (i.e., vapor pressure deficit)], two different sets of hemp cultivars were tested. The first set included eight hemp cultivars known to be grain and fiber cultivars. The second set included six oilseed/CBD cultivars received as transplants from the University of Tennessee's Department of Plant Pathology. The same six oilseed/CBD cultivars were tested in a field experiment during summer 2019.

For the progressive soil-drying study, each set was tested simultaneously across a period of eight weeks. For the TR under high evaporative demand test (growth chamber study), space was limited; therefore, only two cultivars were tested at a time during the course of 12 weeks.

2.1 *Progressive soil-drying study*

The progressive soil-drying tests were conducted in a greenhouse at the West Tennessee AgResearch and Education Center (WTREC) in Jackson, Tennessee, in two sets of experiments. Air temperature was set for 32 day/26 ^{0}C night. During the course of the study, the temperature and relative humidity were measured every five minutes using a universal serial bus (USB) data logger (MicroDaq, Contoocook, NH).

In set 1, the hemp seeds of eight cultivars, viz., "Finola," "Kurd," "Joey," "Felina," 'CFX-2′, "Delores," "Canda," and 'CFX-1′ (for grain and fiber), were initially sown in commercially available planting cubes (Jiffy 50-Cell Tray, Lowe's, Jackson, TN) filled with potting mix (Miracle-Gro). Cultivar 'CFX-1′ had very poor germination; therefore, it was dropped from the experiment. Seeds were planted 1-cm deep, with five seeds cube^{-1} and six cubes per cultivar (total of 30 seeds/cultivar) and kept moist. They were allowed to grow in a reach-in Growth Chamber (PGR15, Conviron, Winnipeg, MB, Canada) for 7 days before transplanting to 5.7-liter pots filled with soil containing 50% sand and

50% silt. In set 2, six additional oilseed/CBD cultivars, viz., "Ha3ze," "Carolina," "T1 Topped (T1 T)," "T1 Not Topped (T1 NT)," "Cherry Topped (Cherry T)," and "Cherry Not Topped (Cherry NT)," which were already in vegetative growth stages, were transplanted into the same silt loam soil 27 days prior to the start date of the experiment.

For set 1, three seedlings per pot were transplanted into nine replications per cultivar (a total of 117 pots). After two weeks, the plant-stand was thinned to one plant per pot. The plants were grown for 22 days under well-watered conditions, watered each day to return to pot capacity, and fertilized once a week using Maxi-Gro (General Hydroponics) nutrient with 10-5-14 (N-P-K). Plants were still in a vegetative stage at this point. If flowers appeared, they were excised and set on the top of the pot to remain in the weight calculation, but prevent any reproductive growth and extra transpiration.

The day before initiation of the soil-drying experiment, all pots for both set 1 and set 2 were watered until there was drainage from the bottom of each pot. Eight plant replications were kept per cultivar. The following morning, each pot was sealed into two double-bagged 15-liter plastic bags (Glad, 15-liter white trash bag, Walmart, Jackson, TN) and secured at the base of the plant with plastic twist ties. A plastic tube (13-mm-diam. × 126-mm-long) was also inserted to facilitate watering. All pots were weighed immediately after bagging, and these measurements constituted the initial weights. Each day, the pots were weighed between 13:00 and 14:00 hours, Central US Time. Daily transpiration was calculated as the difference in weight of each pot on successive days (Shekoofa et al. 2013).

After the second day of these measurements, the plants received water based on their TR for that day. Three pots per cultivar were selected to be "well-watered check," and the other five received the soil-drying treatment ("stressed"). The well-watered plants were watered daily to maintain a weight of no less than 200 g of their initial weight, whereas each of the other five plants was allowed to dry through transpiration. To achieve an extended drying period and to prevent rapid dehydration, the drying pots were watered daily when necessary to maintain daily net water loss to no more than 100 g (Sinclair et al. 2018). This process was continued across a two-week period.

The TR data were subjected to two normalizations, as described by Shekoofa et al. (2013). The first normalization was to minimize the influence of large variations in daily TR across days. The daily transpiration ratio for each drying pot was calculated between its water loss and the average water loss of the three well-watered pots. The second normalization was done to account for variation in transpiration ratio among individual plants. The daily transpiration ratio was divided by the mean transpiration ratio of that same plant during the first three days of the study, when the soil still had high water content. This ratio was identified as the normalized transpiration ratio (NTR), and its value during the

initial wet phase of the dry-down cycle for each plant, by definition, was centered on a value of 1.0.

Data collection for each pot continued until it dried to an NTR ≤0.1, which was defined as the endpoint of transpirable soil water. The total transpirable soil water available to the plant in each pot was calculated as the difference between the initial weight and the final weight of the pot. To track soil drying, FTSW was determined on each day for each pot. The FTSW of each pot was calculated as the difference between daily and final weight, divided by the total transpirable soil water. The use of transpirable soil water as the basis for comparing plant response to soil drying under a range of conditions has been effectively done in a few studies (Ray and Sinclair 1997; Serraj et al. 1999). Data for plants of a cultivar were combined to regress NTR on FTSW.

The data were analyzed using a two linear-segment regression analysis via GraphPad Prism version 8 (GraphPad Software). This regression analysis generated the FTSW threshold for the initiation in the decline in NTR. The 95% confidence interval of the threshold for each cultivar was used to assess statistical differences among cultivars (Sinclair et al. 2018).

2.2 *Growth chamber study under varying evaporative demand*

Two sets of experiments were conducted in this study. First set, eight hemp cultivars; "Finola," "Kurd," "Joey," "Felina," 'CFX-2′, "Delores," "Canda," and 'CFX-1′, were germinated from seeds and grown in a greenhouse at WTREC Jackson, TN. For the second set, four additional oilseed/CBD cultivars; "Carolina," "T1 T," "T1 NT," and "Cherry T" were received as transplants from the University of Tennessee's Department of Plant Pathology.

The plants were grown in pots constructed from polyvinyl chloride pipe (10-cm diameter and 20-cm long). The bottom of each pot was fitted with a flat end cap, in which a small hole was drilled to allow drainage of excess water. A toilet flange was attached to the top of the pot to allow easy attachment of a VPD chamber during measurements (Fletcher, Sinclair, and Allen 2007; Shekoofa et al. 2015). The pots were filled with commercially available Miracle-Gro potting mix. Five seeds per pot were sown. After one week, the plant-stand was thinned to three plants per pot, and after 10 days, it was reduced to one plant per pot. Maxi-Gro (General Hydroponics) with 10-5-14 (N-P-K) was applied once a week. Plants were grown for 28 days under well-watered conditions, with greenhouse temperatures regulated at 32 day/26 ^{0}C night.

Once plants reached the desired growth stage (considering shoot/root ratio), they were moved to a walk-in growth chamber and overwatered until water started dripping from the bottom of each pot. Aluminum foil was placed around the base of each plant, covering the soil to seal in all moisture and prevent soil evaporation. Then the pots were allowed to drain overnight (Shekoofa et al. 2015).

To construct each VPD chamber, a 340-mm diameter food container lid (Cambro Manufacturing, Huntington Beach, CA), with the center cut out, was attached to the toilet flange at the top of the pot. The following morning, the aerial parts of each plant were enclosed in a 21-L clear plastic food container (Cambro Manufacturing, Huntington Beach, CA) by placing the inverted container over the plant and attaching it to the previously installed lid. Each VPD chamber was fitted with a 12-V, 76-mm-diameter computer box fan (Northern Tool and Equipment, Burnsvillle, MN) to continuously stir the air inside the chamber. In addition, a temperature/humidity data logger (MicroDaq, Contoocook, NH) was mounted through the sidewall of each container to measure the chamber environment.

Three levels of VPD were achieved using air flowmeters and controlling the source of the air (ambient or dehumidified). During each of these levels of VPD, TRs were measured. The temperature was kept at a constant 32 ^{0}C comparable to the greenhouse environment. Each day was begun by attaining low VPD (0.5–1.5 kPa), then escalating to medium VPD (1.5–2.5 kPa), and finally reaching high VPD (2.5–3.5 kPa). After achieving the target VPD level, the chamber was allowed to stabilize for 30 min, and then the entire pot chamber system was weighed to the nearest 0.1 g to obtain the initial weight. The plants were then exposed for one hour to each VPD level and then reweighed to measure the final weight. TR at each VPD level was calculated as the difference between the initial weight and the final weight.

Eight plants were studied at a time, allowing two cultivars with four replications. To increase the data points taken on each plant, measurements were done on each plant across two consecutive days. After completing measurements on the first day, the pots were watered and allowed to drain overnight. After completing the second day of measurements, leaves were harvested and total plant leaf area was measured using a leaf area meter (LI-3100, Li-Cor, Lincoln, NE). Finally, the TR for each plant was expressed as water-loss rate divided by plant leaf area.

Data from the two measurement days for all plants of a cultivar in each experiment were combined to perform a two-segment linear regression (Prism 8.0, GraphPad Software Inc., San Diego, CA) for TR vs. VPD. When significantly different ($p < 0.05$) slopes for two segments were not obtained for a cultivar, a simple linear regression was applied to all the data. For those cultivars expressing the two-segment response, the regression analysis yielded the VPD breakpoint between the two linear segments as well as the slope of each segment. The standard error for the VPD breakpoint and the two slopes was an output of the regression analysis. The term breakpoint is used to define the VPD (kPa) point at which the plant begins to close stomata partially and conserve the water remaining in the soil.

2.3 *Field study*

A field study was conducted at a West Tennessee field in Beech Bluff, TN, during summer 2019. Sixteen CBD cultivars were transplanted into the field in the third week of June 2019. All the cultivars were received as transplants from Dr. Eric Walker, Genamera, Springfield, TN. Each cultivar was replicated three times, totaling 48 plots in the field. The trial was established using a randomized complete block design with six replications. Each plot contained two rows of 10 plants spaced 1.2 m apart and rows were spaced 1.8 m apart. Although, because of limited availability of hemp transplants, only six cultivars, viz., "Carolina," "Cherry T," "Cherry NT," "Ha3ze," "T1 T," and "T1 NT," from the controlled-environment studies were available to be considered for the field observation.

During the growing season, soil fertility was managed by applying a per hectare total of 219.5 kg N, 165.5 kg P, and 125.5 kg K preplant. Weeds were managed by cultivation throughout the season. The total rainfall was higher than 30-year average in early June but lower than average in July and August, and much lower than average in September. Temperatures were similar to 30-year average during each month, except September, where temperature was 7–9% higher.

Samples for RWC and specific leaf area (SLA) were taken 63 and 91 days after transplant (DAT) between 11:00 am and 3:00 pm, when the ambient VPD and temperature were up to 4.2 kPa and 37 ^{0}C, respectively. Following sampling methods described by Gonzalez and Gonzalez-Vilar, to measure the RWC and SLA, the youngest but fully expanded leaf from the middle part of the plant was removed and placed in a sealed plastic bag. The sampler would blow into the bag and put samples on ice in the field to ensure that the samples remained fresh and maintained turgidity. Then, each plant was visually assessed for wilting and assigned a score based on severity of wilting, 0 = no wilting and 5 = the plant essentially senesced.

Samples were taken back to the lab where their fresh weight was measured (g) and then immediately after, leaf area was measured with a leaf area meter (LI-3100, Li-Cor, Lincoln, NE). The following day, the leaflet samples were placed in a tub of deionized water for 6–8 h and exposed to 4 ^{0}C in a walk-in cooler before weighing once again for a turgid weight. Leaflet dry weights were measured after placing the leaflets in individual paper envelopes (coin envelopes, Uline.com) and placed in an oven at 80 ^{0}C for 12 h. After all measurements were taken, RWC and SLA were calculated using the following equations:

$$RWC = \frac{fresh\ weight - dry\ weight}{saturated\ weight - dry\ weight} \times 100$$

$$SLA = \frac{leaf\ area}{dry\ weight}$$

Hemp plants were hand-harvested at maturity by cutting the plants at the base, then immediately bucked to remove flower and leaf material from the stem. Flower and leaf material were then dried on mesh screen shelves in temperature- and humidity-controlled rooms. Drying times varied by variety; weights were recorded after drying. All plants in each plot were harvested, but the actual number of plants harvested per plot varied because of differences in plant availability from suppliers and plant survival. Because of this, plot weights were divided by the number of harvested plants and yields are reported on a kg per plant basis. Dried material was sent to Altitude Lab Solutions (Englewood, CO) for analyses of cannabinoid potency using high performance liquid chromatography (HPLC).

The value of max active CBD concentration was determined for each plot and used to calculate CBD yield in kg per plant. Finally, values for hemp physiological parameters were compared among cultivars using one-way ANOVA. Statistical significance was set at $p \leq 0.05$. The treatment means were separated using Tukey's HSD test (SAS Institute, Cary, NC).

3. Results and discussion

3.1 *Progressive soil-drying study*

The two-segment linear regression successfully described the transpiration response rate to soil drying of all hemp cultivars (Table 1). There were substantial differences among the cultivars. The FTSW threshold for cultivar Delores was 0.81 (Table 1). Cultivar Delores had the earliest stomatal closure and highest water saving under soil-drying conditions, whereas Canda with an FTSW threshold of 0.16 (Table 1) had stomatal closure last and maximum water loss early during growth (Table 1). Among all cultivars, the correlation coefficient for the two-segment regression ranged between 0.44 and 0.98 (Table 1). The diverse FTSW thresholds for the tested hemp cultivars support the observations of Devi et al. (2009), in which there was large variation among 17 peanut genotypes in a progressive soil-drying study. Sinclair et al. (2018) reported significant differences in the FTSW threshold in each of the tested peanut populations. Among the 22 North Carolina State University breeding lines tested, FTSW thresholds ranged from 0.38 to 0.69.

Across six hemp cultivars (T1 NT, T1 T, Cherry T, Cherry NT, Carolina, and Ha3ze) tested in the field, there was a negative but significant correlation between low wilting scores (WS) observed in the field and high FTSW thresholds (i.e., early stomatal closure) in the controlled environment

Table 1. Days to reach endpoint of dry-down, fraction of transpirable soil water (FTSW) threshold, and confidence interval in the dry-down and their R^2 for thirteen hemp cultivars transpiration response to progressive soil drying.

Cultivars	Days to endpoint of dry down	FTSW threshold	Confidence interval ($p < 0.05$)	R^2
5.7-liter pots				
Finola	9	0.44	0.34 to 0.54	0.66
Canda	9	0.16	0.10 to 0.21	0.44
Felina	10	0.62	0.40 to 0.85	0.53
Joey	9	0.36	0.24 to 0.48	0.57
Delores	10	0.81	0.71 to 0.90	0.86
CFX-2	10	0.68	0.58 to 0.78	0.81
Kurd	9	0.65	0.56 to 0.73	0.86
Carolina	8	0.32	0.27 to 0.37	0.87
T1 T	11	0.46	0.40 to 0.50	0.92
T1 NT	9	0.34	0.30 to 0.36	0.97
Cherry T	8	0.28	0.25 to 0.29	0.98
Ha3ze	12	0.55	0.38 to 0.68	0.78
CFX-1 †	-	-	-	-
15-liter pots‡				
Ha3ze	23	0.56	0.50 to 0.62	0.78
Cherry NT	22	0.44	0.37 to 0.52	0.73

†Dropped from experiment due to unsuccessful germination in greenhouse.
‡The measurements of TR for these two cultivars were done under the same experimental condition but in 15-liter pots.

(Figure 3). These results were indicative of a correspondence between the FTSW threshold for TR decrease because of early stomatal closure and expression of soil water conservation in the field. Furthermore, five cultivars, viz., T1 NT, T1 T, Cherry NT, Carolina, and Ha3ze, had significantly higher RWC (Figure 2) in the field compared to cultivar Cherry T that had a FTSW threshold of 0.28 (Table 1). There were no significant differences among the tested cultivars for SLA under field conditions (data not shown).

3.2 *Transpiration response to high evaporative demand*

Seven of the 12 hemp cultivars expressed the TR_{lim} trait, with their VPD breakpoints (BPs) ranging from 1.2 to 2.6 kPa when the TRs were measured under varying VPD rates (0.5–3.5 kPa). The other four cultivars had a linear TR response under dry air (high VPD, >2.5 kPa) (Table 2). The R^2 for the hybrids described by the two-segment regression ranged from 0.43 to 0.87 (Table 2). Cultivar Canda that had the lowest FTSW threshold in the progressive soil-drying study also had a high VPD PB of 2.5 kPa; whereas, cultivars Delores and Kurd, with high FTSW thresholds, both had a VPD BP of 2.1 kPa (Table 2). The seven cultivars that expressed the TR_{lim} trait showed no correlation between their VPD breakpoints (BPs) in the controlled environment and wilting score under the field condition. Therefore, temperature differences used in the controlled environment (32^0C) during the experiment and experienced in the field (37^0C) may be a particularly

Table 2. Results from two-segment linear regression for the hemp cultivars that were found to fit this model and linear regression for the remaining cultivars. The table contains Slope 1 (± SE), breakpoint (BP ± SE), confidence limit of the BP, Slope 2 (± SE), and their $R^{2.}$

Cultivars	*n*	Slope 1± SE	Slope 2± SE	BP (X_0) ±SE	Confidence limit of X_0	R^2
Finola	22	60.97 ± 8.91	−78.95 ± 64.89	2.6 ± 0.26	2.03 to 3.16	0.74
Canda	24	41.77 ± 14.07	−109.5 ± 44.84	2.5 ± 0.15	2.15 to 2.78	0.46
Felina	28	67.97 ± 10.85	Linear	-	-	0.60
Joey	24	80.46 ± 8.83	Linear	-	-	0.79
Delores	24	59.96 ± 18.82	−85.17 ± 52.48	2.1 ± 0.21	1.94 to 2.85	0.43
CFX-1	24	41.26 ± 14.66	18.17 ± 10.44	2.0 ± 0.85	0.22 to 3.77	0.79
CFX-2	28	57.04 ± 7.34	Linear	-	-	0.70
Kurd	22	94.21 ± 12.54	21.86 ± 26.57	2.1 ± 0.30	1.45 to 2.74	0.87
Carolina	24	31.07 ± 4.01	Linear	-	-	0.73
Cherry T	24	25.99 ± 3.43	Linear	-	-	0.72
T1 T	24	52.66 ± 10.93	1.38 ± 6.02	1.6 ± 0.17	1.19 to 1.91	0.78
T1 NT	24	105.4 ± 26.77	4.98 ± 3.91	1.2 ± 0.08	1.02 to 1.36	0.75

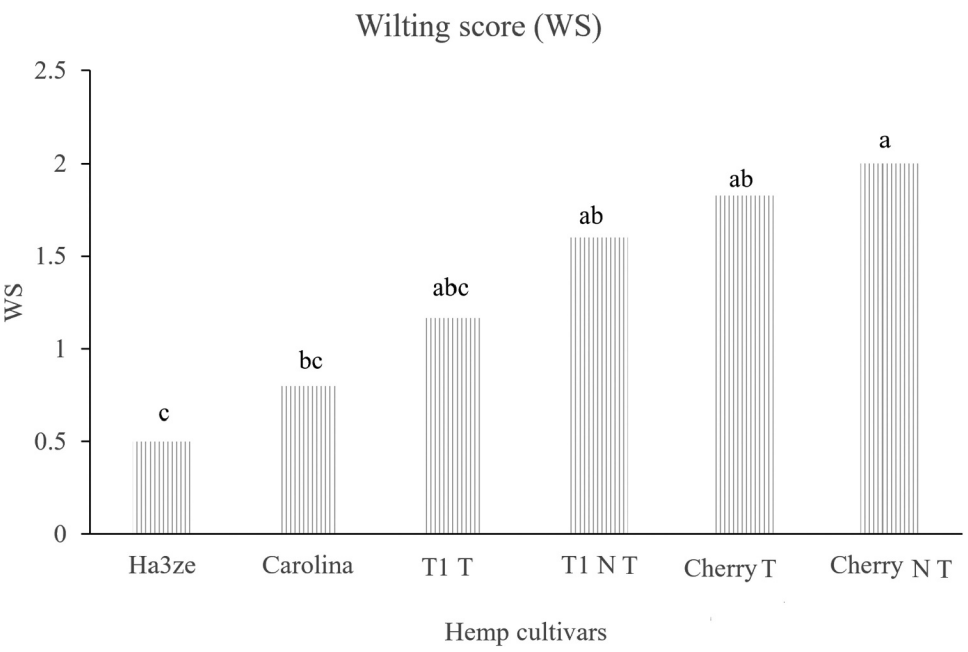


Figure 1. The graph of wilting score (WS) responses for six hemp cultivars under rainfed conditions in the field trial.

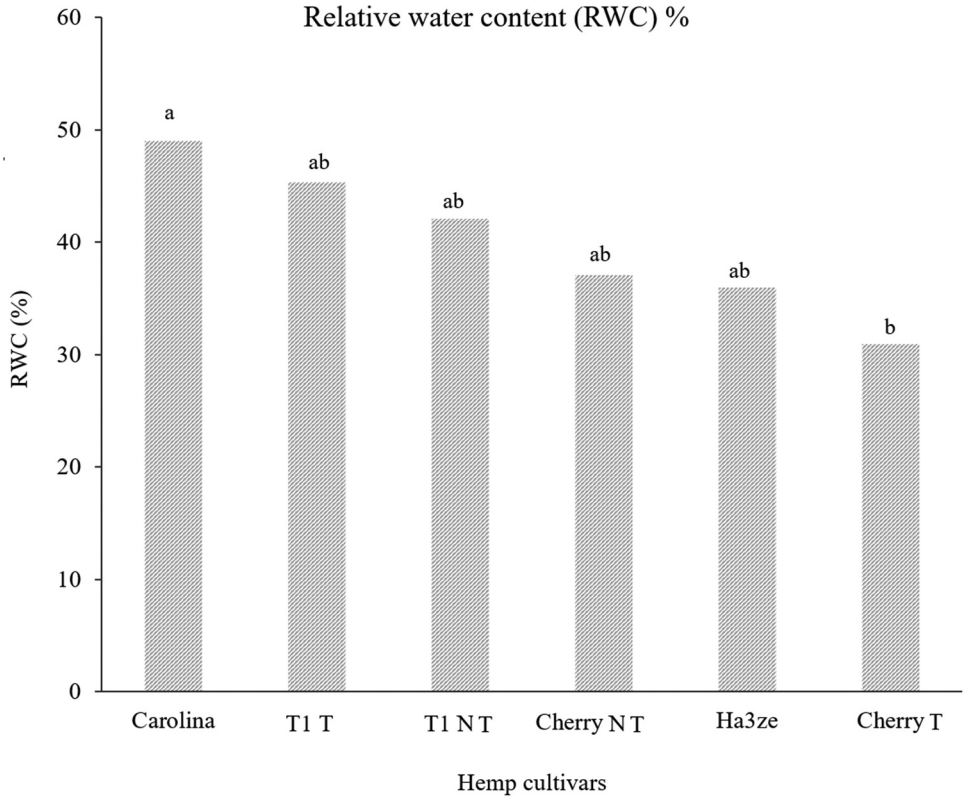


Figure 2. The relative water content (RWC) rates for six hemp cultivars under rainfed conditions in the field trial.

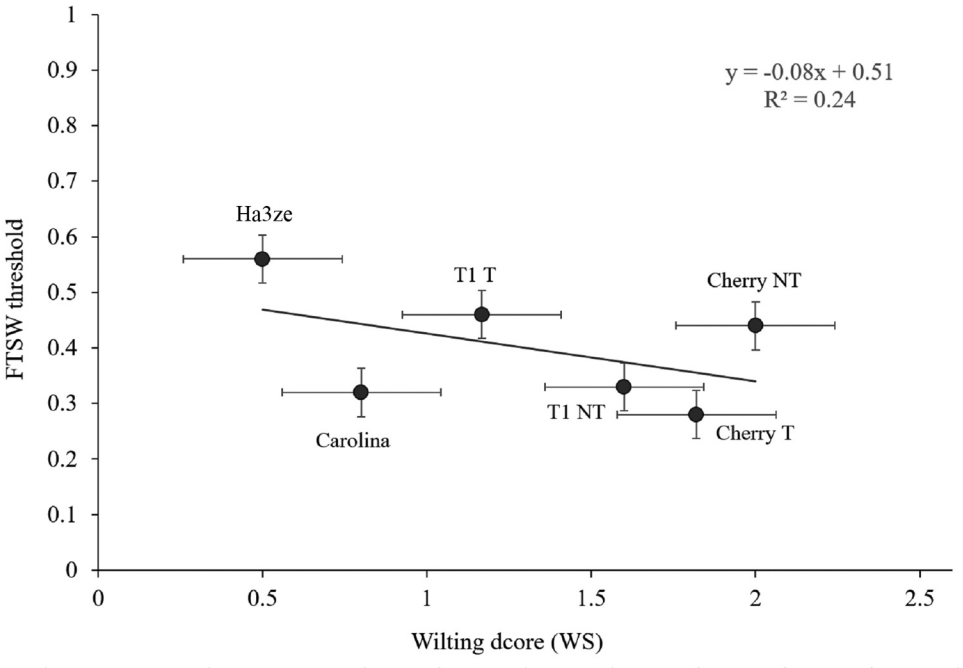


Figure 3. Correlation of fraction of transpirable soil water (FTSW) threshold in the controlled environment with wilting score (WS) in the field for hemp cultivars. Error bars represent standard error of the mean.

important point to consider. This result contrasts with the results from previous comparisons of the TR_{lim} trait between controlled-chamber measurements and field measurements in soybean (Gilbert et al. 2011) and sorghum (Shekoofa, Balota, and Sinclair 2014).

3.3 *Field study*

The natural daily range of evaporative demand in the field allowed measurements to be made under ambient VPD up to 4.2 kPa and temperature up to 37^{0}C while plants experienced rainfed conditions the entire season. The increase in midday temperature resulted in progressively higher evaporative demand through the day. Therefore, all visual wilting scores were recorded between 11:00 a.m. and 1:00 p.m. Cultivars Ha3ze, Carolina, and T1 T had the lowest wilting scores (Figure 1). Furthermore, cultivars Ha3ze, Carolina, and T1 T had significantly higher RWC rates compared to Cherry T (Figure 2). Figure 3 depicts a negative correlation (p = 0.05) between FTSW threshold in the controlled environment and wilting scores recoded in the field for the tested hemp cultivars (Figure 3).

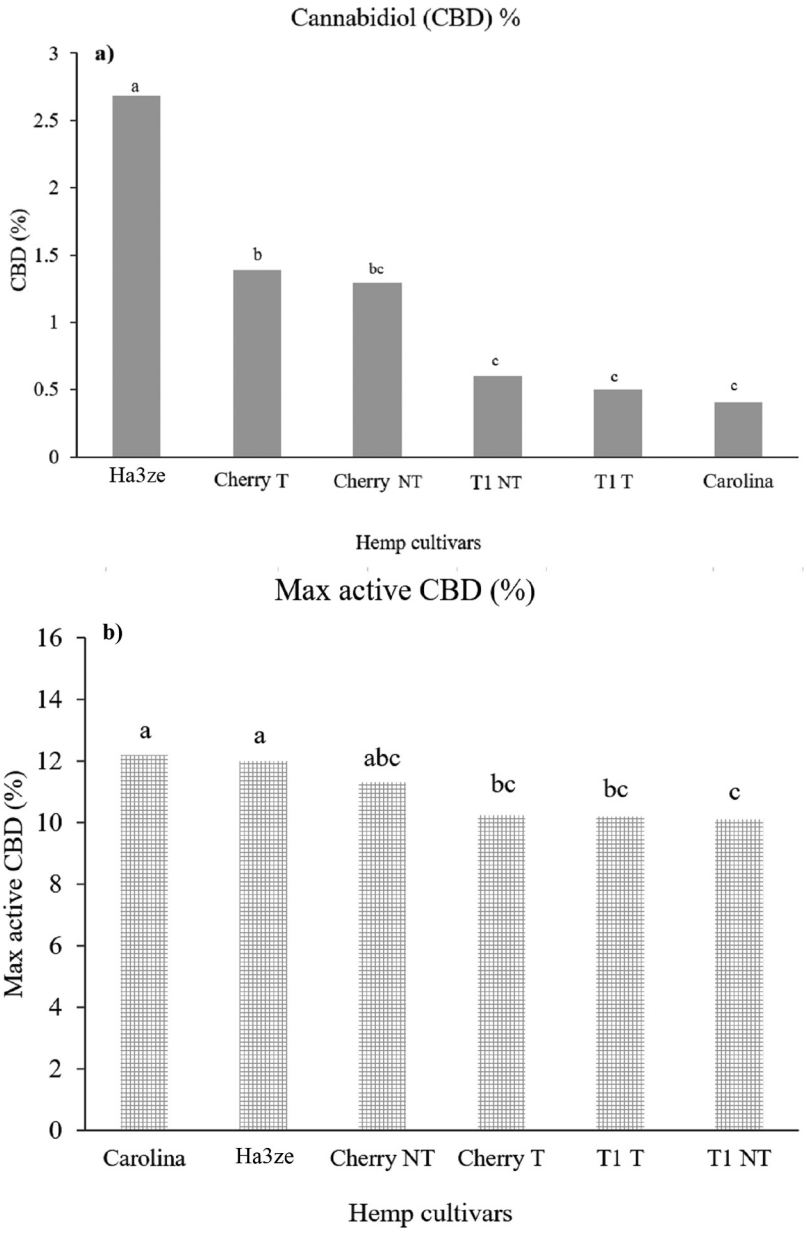


Figure 4. Hemp cultivars cannabidiol (CBD) % (a) and max active CBD % (b) under the field condition.

Selected hemp cultivars were evaluated in the field for their CBD as part of ongoing Tennessee Hemp Variety trials. These tested hemp cultivars were grown to maturity, harvested and dried for CBD processing. Cultivar Ha3ze had the highest CBD content (2.68%) among cultivars (Figure 4a). The max active CBD percentage was significantly higher for Ha3ze cultivar compared

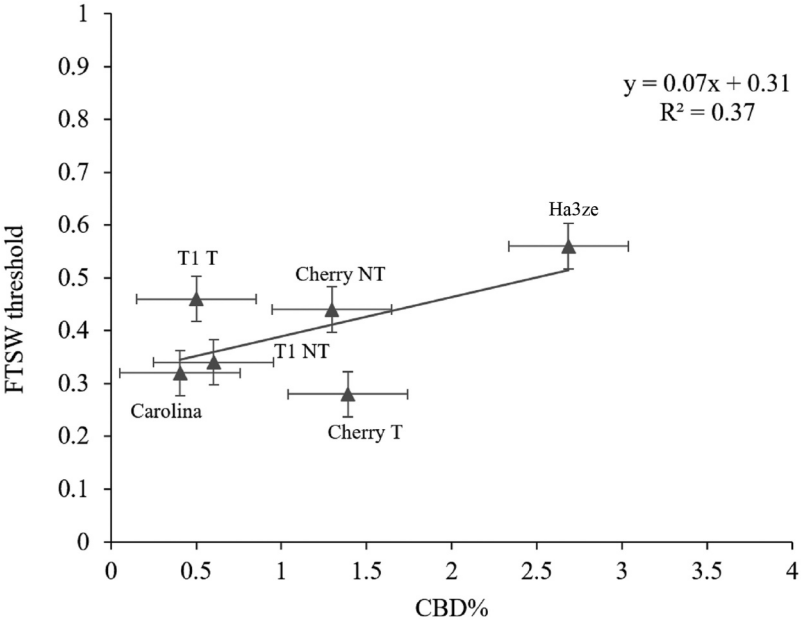


Figure 5. Correlation of fraction of transpirable soil water (FTSW) threshold in the controlled environment with cannabidiol (CBD) % in the field for hemp cultivars. Error bars represent standard error of the mean.

to T1 NT, T1 T, and Cherry T cultivars (Figure 4(b)). There was a positive correlation (p = 0.05) between FTSW threshold and CBD percentage for the tested hemp cultivars (Figure 5). Among all six tested cultivars, Ha3zE had the highest FTSW threshold and earliest stomatal closure during progressive soil drying (Table 1), and it had the highest CBD percentage compared to other cultivars (Figure 5).

There was a positive and significant correlation between FTSW thresholds and percentage of CBD for six cultivars (Ha3ze, T1 NT, T1 T, Carolina, Cherry T, and Cherry NT) tested in the field (Figure 5). Those cultivars that had higher FTSW threshold under water deficit stress (i.e., progressive soil drying study) not only produced higher percentage of CBD but also produced more max active CBD percentage than those hemp cultivars with low FTSW threshold (Table 1; Figure 5). Because hemp value is determined by both biomass yield and the percentage of CBD, selecting cultivars based on high CBD yield can help maximize farmers' profit under rainfed conditions.

Among the six cultivars evaluated in the field, Ha3ze that ranked the highest for the FTSW threshold under progressive soil drying had the highest percentage of CBD under ambient conditions. Cultivar Ha3ze produced 80% higher CBD compared to T1 NT, T1 T, and Carolina.

5. Conclusion

Soil water conservation for hemp can be a very important factor; therefore, identifying drought-tolerance traits (i.e., early stomatal closure under soil drying and TR_{lim}) is critical for improving hemp production. This research demonstrates that variation among hemp cultivars for the drought-tolerance traits should help in identifying superior cultivars for rainfed environments. The expression of the drought-tolerance traits was generally characterized similarly under both controlled environments and field conditions. Consequently, the results of this study that showed stability in the expression of the early stomatal closure in progressive soil-drying conditions should accelerate future efforts toward comparing germplasm for the water-saving trait under a range of conditions.

Disclosure statement

No potential conflict of interest was reported by the authors.

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